

Advanced Applied Econometrics

Static discrete choice with market-level data

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19 June 2026, DIW

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- Next class on June 26
 - Single agent dynamic discrete choice
 - Rust (1987): "Optimal replacement of GMC bus engines: an empirical model of Harold Zurcher"
 - Graded problem set will be distributed this week. Due by Monday, July 6, 23:59.

Plan for today

- Recap BLP: The Random Coefficient Logit Model of Demand
- Go through BLP code
- Discuss addition of supply-side moments

- Individual choice model using market-level data with
 - horizontal (ϵ, μ) and vertical (δ, ζ) product differentiation
 - (price) endogeneity
 - Isolate mean utility to estimate β coefficients by linear IV estimation
 - unobserved consumer heterogeneity
 - flexible substitution patterns
 - In homogenous logit, only market shares matter
 - In heterogenous logit, closeness in characteristics space $\rightarrow$ product differentiation matters
 - static pricing game on supply side can improve identification
 - quasilinear preferences allow computing changes in consumer welfare

BLP widely used but problems

- Econometric
 - Identifying unobserved heterogeneity with aggregate data can be hard in practice: Petrin (JPE 2002), BLP (JPE 2004) add microdata
 - Weak instruments due to lack of cost shifters: Armstrong (Ecta 2016), Reynaert and Verboven (JoE 2014), Berry and Haile (2014), Gandhi and Houde (2019), handbook chapter Gandhi and Nevo (2021)
 - "Logit" assumption / Welfare analysis with many products, entry/exit: Akerberg and Rysman (Rand 2005), Berry and Pakes (IER 2007)
 - Measurement error in market shares (small T , large J): Gandhi, Lu, and Shi (QE 2023)

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- Many solutions implemented in Conlon and Gortmaker (2020): pyblp

Identification and instruments

- exogenous variation in observed characteristics
- choice set variation (over multiple markets)
- variation in preferences

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- choice set variation (over multiple markets)
- variation in preferences
- formal positive nonparametric identification results by Berry and Haile (Ecta 2014, ARE 2016), Fox and Gandhi (Rand 2016), Fox, Kim, Ryan, and Bajari (JoE 2012)
→ functional form and distributional assumptions not necessary, standard IV

- In the problem set, we estimate $\{\alpha, \beta\}$ on simulated data in three ways:
 - OLS: biased
 - IV with BLP instruments (rivals' characteristics) and x_j^2
 - IV adding cost shifters w_j
 - Two issues we glossed over:
 - Why are these valid instruments, and which ones actually carry identifying power?
 - With random coefficients we also need instruments to identify σ
- Today: some more on exclusion restrictions, weak-instrument problem, optimal instruments

Two endogeneity problems

- Nonlinear GMM on $E[\tilde{\zeta}_j \mid x_j, z_j] = 0$. Any function $f(x_j, z_j)$ is then a valid instrument:

$$E[\tilde{\zeta}_j f(x_j, z_j)] = 0$$

- So we may use $x, x^2, x^3, \dots$, interactions $x \cdot z$, etc. – how to choose: optimal IV
- There are two distinct endogeneity problems:
 - Price p_j : correlated with $\tilde{\zeta}_j$, e.g. through costs
 - Nonlinear parameter σ : with heterogeneity, substitution depends on full vector $\tilde{\zeta}$

→ Need instruments for both α and σ

Why does σ need an instrument?

- σ identified by substitution: how does s_j respond to neighbours in characteristic space
- Linearise the model (Salanié–Wolak 2019):

$$\ln \frac{s_j}{s_0} \approx x_j \beta + \sigma^2 a_j + \zeta_j, \quad \text{with } a_j = \left(\frac{x_j}{2} - \sum_k s_k x_k \right) x_j$$

- Regressor a_j is built from s_k , which depends on whole vector ζ (incl. ζ_j)
- $\Rightarrow \text{Cov}(a_j, \zeta_j) \neq 0$: the “ σ regressor” is endogenous
- Fix: instrument for a_j , e.g. using exogenous $x_{-j} \rightarrow$ differentiation IVs (next); valid since $E[\zeta_j \mid x] = 0$
 - Without an IV, σ is identified by functional form (Berry and Haile 2014)

Exclusion restrictions (Berry and Haile 2014)

$$\delta_j = x_j\beta - \alpha p_j + \xi_j \quad (\text{demand})$$

$$\ln(p_j - \eta_j(\theta, p, s)) = h(x_j, w_j; \gamma) + \omega_j \quad (\text{supply / markups})$$

First place to look for instruments: something in the other equation

- w_j – shifts supply (marginal cost) but not demand $\rightarrow$ instrument for p_{jt}
- v_j – a demand shifter excluded from cost (e.g. an excluded characteristic)
- Cross-market variation in the number/strength of competitors shifts equilibrium markups

Alternative when cost shifters are scarce: MacKay and Miller (2023) use $\text{Cov}(\xi_j, \omega_j) = 0$

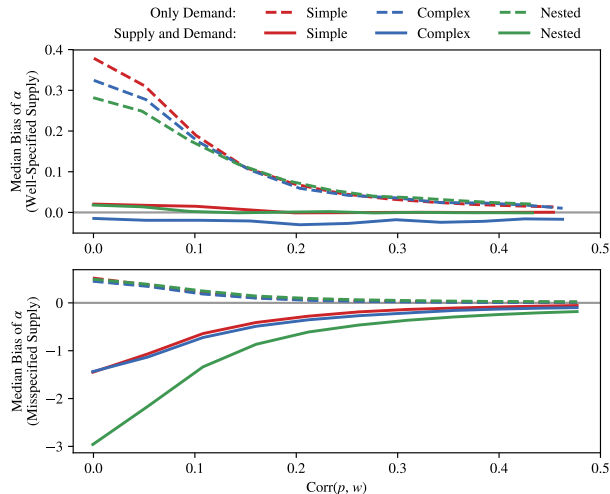
BLP instruments and why they can be weak

- Characteristics of *other* products, since markups fall when close substitutes are present
 - own-firm rivals $\sum_{k \in \mathcal{F}_j} x_k$; other-firm rivals $\sum_{k \notin \mathcal{F}_j} x_k$ (plus 1, x_j , interactions)
- Armstrong (2016): as $J \rightarrow \infty$ the markup term collapses to a constant,

$$\frac{s_{jt}}{|\partial s_j / \partial p_j|} = \frac{1}{\alpha} \frac{1}{1 - s_j} \longrightarrow \frac{1}{\alpha}$$

- Hard to instrument for a constant: averages of x_{-j} are an especially poor choice
 - Power survives mainly through variation in #products per market / per firm
- cost shifters needed

Cost shifters matter



- Bias in $\hat{\alpha}$ as instrument strength varies (Conlon and Gortmaker 2020)
- BLP instruments: weak, large bias
- Adding cost shifters $\rightarrow$ strong first stage, bias vanishes

$\rightarrow$ Always inspect $E[p \mid x, z]$ first

Differentiation instruments (Gandhi and Houde 2019)

- To identify σ we want instruments measuring how differentiated j is locally
- Instead of *levels/averages* of rivals' x , use distances $d_j^k = |x_k - x_j|$:

$$\text{DIV}_1 : \sum_{k \in \mathcal{F}_j} (d_j^k)^2, \sum_{k \notin \mathcal{F}_j} (d_j^k)^2 \quad \text{DIV}_2 : \sum_k \mathbb{I}[d_j^k < c]$$

- Intuition: a product with many close neighbors has very different substitution than an isolated one – variation to pin down σ

Optimal instruments (Chamberlain 1987, Newey 1990)

- Asymptotic GMM variance depends on $(G'\Omega^{-1}G)^{-1}$ with $G = E\left[\frac{\partial \tilde{\zeta}_j}{\partial \theta} \mid z\right]$
- Optimal instruments set $f(z)$ to the expected Jacobian of the moments, scaled by Ω^{-1} :

$$Z_{jt}^{opt} = E\left[\frac{\partial \tilde{\zeta}_j}{\partial \theta} \mid z_t\right] \Omega_t^{-1}$$

- Infeasible: depend on solution θ_0
- Since any $f(x, z)$ is valid, higher-order terms $x, x^2, \dots$ are an attempt to approximate this

- BLP (1999) feasible approximation:
 1. Fix $\hat{\theta}$, draw (ξ^*, ω^*) (lazy: set = 0)
 2. Solve FOCs for $\hat{p}_t$
 3. Solve shares $s_t(\hat{p}_t, \hat{\theta})$
 4. Compute the Jacobian, average over draws
- Reynaert and Verboven (2014): easier, useful solution if assuming $p \approx mc$
- Either way, the first stage $E[p \mid x, z]$ is the most important (good instruments in the data).

- Always check the first stage $E[p \mid x, z]$ before going into full model
- BLP instruments alone are often weak
- Use differentiation IVs
- In practice: `results.compute_optimal_instruments()` in `pyblp`
- Add supply side – if assumptions fine for counterfactuals, may as well use for identification

Supply Side

Supply-side equilibrium model

- How are prices determined in oligopolistic markets with differentiated products?
- → counterfactual policy analysis
- Increased efficiency of demand estimates by adding structure on the supply side
- Assume prices are result of firms' optimization problem
- Use information from first-order conditions for estimation
- Single-product vs. multi-product firms
- Specify cost function and equilibrium notion

Supply-side equilibrium model

Berry (1994) single-product firms

- Profits of firm j are given by $\Pi_j(p) = p_j q_j(p) - C_j(q_j(p))$
- Conduct assumption: compete à la Nash in prices
- The first-order condition is:

$$q_j + (p_j - mc_j) \frac{\partial q_j}{\partial p_j} = 0$$

- which, by rewriting, implies the *markup* term

$$b_j = p_j - mc_j = \frac{-q_j}{\partial q_j / \partial p_j}$$

Supply-side equilibrium model

- For simplicity, assume that marginal costs are constant and take the form:

$$\ln(mc_j) = w_j\gamma + \omega_j \quad (1)$$

- The equation to be estimated is therefore

$$\ln(p_j - b(p_j, x_j, \theta_1, \theta_2)) = w_j\gamma + \omega_j$$

where w_j are observed and ω_j unobserved product characteristics

- The markup $b(p, x, \theta_1, \theta_2) = \Delta(p, x, \theta_1, \theta_2)^{-1}s(p, x, \theta_1, \theta_2)$ depends on demand parameter and, through the equilibrium price, on the unobserved cost component ω

- Estimate demand and supply simultaneously. Define additional instruments $z_j = (x_j, w_j)$.
- Additional moment conditions

$$E[\xi_j|z] = E[\omega_j|z] = 0$$

where $\omega = mc - [x, w]\gamma$

BLP (1995) look at the more general case of a multiproduct firm f producing a subset $\mathfrak{S}_f$ of the J products in the market

- Profits of firm f are given by:

$$\Pi_f = \sum_{j \in \mathfrak{S}_f} (p_j - mc_j) Ms_j(p, x, \theta_1, \theta_2)$$

- The usual assumption is that firms compete à la Nash in prices

Multiproduct firms

- The first-order condition is:

$$s_j(p, x, \theta_1, \theta_2) + \sum_{r \in \mathfrak{S}_f} (p_r - mc_r) \frac{\partial s_r(p, x, \theta_1, \theta_2)}{\partial p_j} = 0$$

- which can be rewritten in vector notation as:

$$p_j = mc_j + \Delta(p, x, \theta_1, \theta_2)^{-1} s(p, x, \theta_1, \theta_2)$$

where the (j, r) element of the J by J matrix $\Delta(p, x, \theta_1, \theta_2)$ is

$$\Delta_{jr} \begin{cases} 1 \times \frac{-\partial s_r}{\partial p_j} = \frac{-\partial s_r}{\partial p_j} & \text{if } r \text{ and } j \text{ are produced by the same firm} \\ 0 \times \frac{-\partial s_r}{\partial p_j} = 0 & \text{otherwise} \end{cases}$$

Multiproduct firms: note on conduct

- Rewrite profit maximization problem:

$$\Pi_f = \sum_{j \in \mathfrak{S}_f} (p_j - mc_j) Ms_j(p, x, \theta_1, \theta_2) + \kappa_{fg} \sum_{j \in \mathfrak{S}_g} (p_j - mc_j) Ms_j(p, x, \theta_1, \theta_2)$$

- The first-order condition is then:

$$s_j(p, x, \theta_1, \theta_2) + \sum_{r \in (\mathfrak{S}_f, \mathfrak{S}_g)} \kappa_{fg} (p_r - mc_r) \frac{\partial s_r(p, x, \theta_1, \theta_2)}{\partial p_j} = 0$$

- Instead of 0's and 1's, now $\kappa_{fg} \in [0, 1]$ represents how much firm f cares about profits of g
- In reality, $\kappa_{fg} \in (0, 1)$, but evidence that $\kappa_{fg} > 0$ not necessarily “anti-competitive” – just deviation from static Bertrand pricing
- Berry and Haile (2014): κ_{fg} can be identified using IV that shifts demand but not supply (“rotate demand”)

What does a demand system actually deliver?

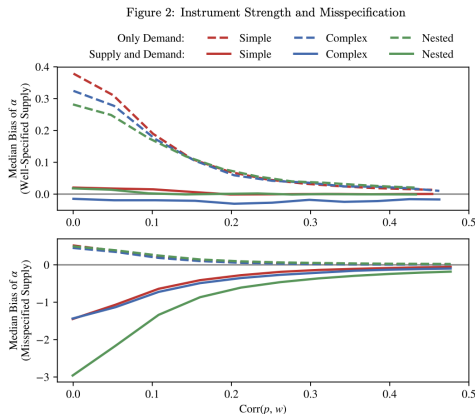
From $\hat{\theta}_2$ everything else follows:

$$mc = p - \underbrace{(\mathcal{H} \odot \Delta(p))^{-1} s(p)}_{\text{markup } b(p, x, \theta_2)}$$

- Own-price elasticities $\epsilon_{jj}(p, \theta_2)$
- Substitution patterns – all transforms of the same object $\Delta(p, \theta_2)$:
 - cross-elasticities ϵ_{jk} , diversion ratios $D_{jk} = \frac{\partial q_k / \partial p_j}{|\partial q_j / \partial p_j|}$
 - diversion to the outside good D_{j0}

→ Counterfactuals (mergers, taxes) crucially depend on these

Does imposing supply help? (Conlon and Gortmaker 2020)



Each plot documents how bias of the linear parameter on price, α , decreases with the strength of the cost shifter w_{jt} , which is included as a demand-side instrument. To weaken or strengthen the instrument, we vary its supply-side parameter from $\gamma_w = 0$ to $\gamma_w = 1$, and report the correlation this induces between w_{jt} and prices p_{jt} . Reported bias values are medians across 1,000 different simulations. The top plot reports results for the simulation configurations described in Section 5. In the bottom plot, we simulate data according to perfect competition (i.e., prices are set equal to marginal costs instead of those that satisfy Bertrand-Nash first order conditions), but continue to estimate the model under the assumption of imperfect competition. For all problems, we use the “approximate” version of the feasible optimal instruments and a Gauss-Hermite product rule that exactly integrates polynomials of degree 17 or less.

- A well-specified supply side sharpens θ_2 (especially price)
- Helps if you have structural information to impose on marginal cost
- Enforce economic constraints, e.g.
 $mc_{jt} > 0$
- But a wrong conduct assumption $\mathcal{H}_t$ injects misspecification

Micro moments

Why aggregate data can underidentify (Π, Σ)

- Heterogeneity parameters need cross-market variation in demographics and choice sets
 - If demographic distribution (m^y, v^y) barely moves across markets, then $m^y x_j, v^y a_j$ are nearly collinear with x_j, a_j
 - Π (observed heterogeneity) vs. Σ (unobserved heterogeneity) separated only by functional form
- exploit within-market micro data linking demographics to individual choices (Petrin 2002; BLP 2004; Conlon and Gortmaker 2024)

- Same model, demographics y_{it} now enter explicitly:

$$u_{ij} = \delta_j + \underbrace{x_j'(\Pi y_i + \Sigma v_i)}_{\mu_{ij}} + \varepsilon_{ij}, \quad s_{ij} = \frac{\exp(\delta_j + \mu_{ij})}{1 + \sum_k \exp(\delta_k + \mu_{ik})}$$

- Micro data = observations linking a consumer's demographics y to the product j they chose
 - surveys of recent buyers, loyalty cards, NielsenIQ panels, second-choice data
- Identifies the part of μ_{ij} that aggregate shares cannot

Estimation: stack aggregate and micro moments

- Aggregate moments (from inversion): $g_A(\theta) = \frac{1}{N_A} \sum_j \hat{\xi}_j(\theta) z_j$
- Micro moments: $g_M(\theta) = \bar{v} - \mathbb{E}_A[\bar{v} \mid \theta]$
- One GMM objective, stacked:

$$\hat{\theta} = \arg \min_{\theta} \begin{pmatrix} g_A(\theta) \\ g_M(\theta) \end{pmatrix}' W \begin{pmatrix} g_A(\theta) \\ g_M(\theta) \end{pmatrix}$$

- Caveat: aggregate and micro data drawn from different populations can be problematic

Takeaway: micro-BLP

- Aggregate BLP is the foundation
- Micro moments are now commonly used if micro data exist
- They identify substitution directly rather than through functional form → better finite-sample behavior
- Same GMM routine, implemented in `pyblp`

BLP Code

Nested fixed point algorithm (BLP 1995)

Functions to be called (directly and indirectly) from main script

- Contraction mapping: $\delta^{h+1} = \delta^h + \ln(S) - \ln(s(\delta^h, \sigma))$
- Market shares: $s_{jt}(\delta_t, \sigma) \approx \sum_{i=1}^n \phi_i \frac{\exp(\delta_{jt} + \mu_{jt}(v_i))}{1 + \sum_{l=1}^J \exp(\delta_{lt} + \mu_{lt}(v_l))}$
- GMM objective function,
minimization problem based on moment conditions $E[g_{jt}(z_{jt})\tilde{\zeta}_{jt}] = 0$:

$$\min_{\theta} \tilde{\zeta}(\theta)' g(z)' A g(z) \tilde{\zeta}(\theta)$$

- For simplicity, no analytical gradients. Need to be used in practice!

Nested fixed point algorithm (BLP 1995)

For data generation, functions needed for

- Equilibrium prices, using FOC: $0 = c_t - p_t - \Delta_t^{-1} s_t$
- Market share derivatives Δ_t :

$$\sum_{i=1}^n \phi_i [-\alpha s_{ijt}(1 - s_{ijt})] \quad \forall j = k, \quad \sum_{i=1}^n \phi_i [\alpha s_{ijt} s_{ikt}] \quad \forall j \neq k$$

Stochastic / Monte Carlo

- “Monte Carlo is the art of approximating an expectation by the sample mean of a function of simulated random variables” - Statistical Genetics lecture notes, UC Berkeley.
- Pseudo-random - standard random number generator
- Quasi-random - more uniform coverage, e.g. Halton, modified Latin hypercube sampling

Non-stochastic / Quadrature

- Gaussian Hermite product rule
 - Difficult for high-dimensional distributions and complicated integrands
- Sparse grid integration (Heiss and Winschel, JoE 2008)
 - Subset of nodes from product rule
- Simulation bias in MSL and MSS (ln in simulated ln $P_n(\theta)$ is a nonlinear transformation), not in MSM